


# DATABASE LAB

*LAB TASK#3*

NAME: Mashaal Ishtiaq

ROLL NO: 22P-9326

SECTION: BCS-6B

# Q1

```
-- 22P-9326
SELECT * FROM employees WHERE first_name || last_name LIKE '%A%';
```

Query Result

All Rows Fetched: 14 in 0 seconds

| | EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 103 | Alexander | Hunold | AHUNOLD | 590.423.4567 | 03-JAN-06 | IT_PROG | 9000 | (null) | 102 | 60 |
| 2 | 105 | David | Austin | DAUSTIN | 590.423.4569 | 25-JUN-05 | IT_PROG | 4800 | (null) | 103 | 60 |
| 3 | 115 | Alexander | Khoo | AKHOO | 515.127.4562 | 18-MAY-03 | PU_CLERK | 3100 | (null) | 114 | 30 |
| 4 | 121 | Adam | Fripp | AFRIPP | 650.123.2234 | 10-APR-05 | ST_MAN | 8200 | (null) | 100 | 50 |
| 5 | 130 | Mozhe | Atkinson | MATKINSO | 650.124.6234 | 30-OCT-05 | ST_CLERK | 2800 | (null) | 121 | 50 |
| 6 | 147 | Alberto | Errazuriz | AERRAZUR | 011.44.1344.429278 | 10-MAR-05 | SA_MAN | 12000 | 0.3 | 100 | 80 |
| 7 | 158 | Allan | McEwen | AMCEWEN | 011.44.1345.829268 | 01-AUG-04 | SA_REP | 9000 | 0.35 | 146 | 80 |
| 8 | 166 | Sundar | Ande | SANDE | 011.44.1346.629268 | 24-MAR-08 | SA_REP | 6400 | 0.1 | 147 | 80 |
| 9 | 167 | Amit | Banda | ABANDA | 011.44.1346.729268 | 21-APR-08 | SA_REP | 6200 | 0.1 | 147 | 80 |
| 10 | 174 | Ellen | Abel | EABEL | 011.44.1644.429267 | 11-MAY-04 | SA_REP | 11000 | 0.3 | 149 | 80 |
| 11 | 175 | Alyssa | Hutton | AHUTTON | 011.44.1644.429266 | 19-MAR-05 | SA_REP | 8800 | 0.25 | 149 | 80 |
| 12 | 185 | Alexis | Bull | ABULL | 650.509.2876 | 20-FEB-05 | SH_CLERK | 4100 | (null) | 121 | 50 |
| 13 | 187 | Anthony | Cabrio | ACABRIO | 650.509.4876 | 07-FEB-07 | SH_CLERK | 3000 | (null) | 121 | 50 |
| 14 | 196 | Alana | Walsh | AWALSH | 650.507.9811 | 24-APR-06 | SH_CLERK | 3100 | (null) | 124 | 50 |

```
-- 22P-9326
SELECT * FROM employees WHERE first_name || last_name LIKE '%_e';

-- 22P-9326
```

Query Result

All Rows Fetched: 5 in 0 seconds

| | EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 128 | Steven | Markle | SMARKLE | 650.124.1434 | 08-MAR-08 | ST_CLERK | 2200 | (null) | 120 | 50 |
| 2 | 135 | Ki | Gee | KGEE | 650.127.1734 | 12-DEC-07 | ST_CLERK | 2400 | (null) | 122 | 50 |
| 3 | 163 | Danielle | Greene | DGREENE | 011.44.1346.229268 | 19-MAR-07 | SA_REP | 9500 | 0.15 | 147 | 80 |
| 4 | 165 | David | Lee | DLEE | 011.44.1346.529268 | 23-FEB-08 | SA_REP | 6800 | 0.1 | 147 | 80 |
| 5 | 166 | Sundar | Ande | SANDE | 011.44.1346.629268 | 24-MAR-08 | SA_REP | 6400 | 0.1 | 147 | 80 |

```
-- 22P-9326
SELECT * FROM employees WHERE first_name || last_name LIKE 'l%';
```

Query Result

All Rows Fetched: 0 in 0.016 seconds

| | EMPLOYE... | FIRST_NA... | LAST_NAME | EMAIL | PHONE_N... | HIRE_DATE | JOB_ID | SALARY | COMMISS... | MANAGER... | DEPARTM... |
|---|---|---|---|---|---|---|---|---|---|---|---|

```
-- 22P-9326
SELECT * FROM employees WHERE first_name || last_name LIKE '_l_s%';
```

Query Result

All Rows Fetched: 2 in 0 seconds

| | EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 149 | Eleni | Zlotkey | EZLOTKEY | 011.44.1344.429018 | 29-JAN-08 | SA_MAN | 10500 | 0.2 | 100 | 80 |
| 2 | 196 | Alana | Walsh | AWALSH | 650.507.9811 | 24-APR-06 | SH_CLERK | 3100 | (null) | 124 | 50 |

# Q2

```
--22p-9326
create user p229326_Lab03 identified by db1234;
```

Script Output

Task completed in 0.062 seconds

```
User P229326_LAB03 created.
```

```
--22p9326
Grant all PRIVILEGES to p229326_Lab03;
```

Script Output

Task completed in 0.047 seconds

```
Grant succeeded.
```

Q3

```
--22p-9326
CREATE TABLE Employees (
    Employee_id INT,
    Full_Name VARCHAR2(100),
    Salary NUMBER,
    Department_id INT,
    Start_Date DATE,
    End_Date DATE,
    Married VARCHAR2(3),
    Phone_No VARCHAR2(11)
);
```

Script Output | Task completed in 0.047 seconds

```
Table EMPLOYEES created.
```

Q4

```
--22p-9326
CREATE TABLE Departments (
    Department_id INT,
    Department_name VARCHAR2(100),
    Department_code VARCHAR2(10),
    Date_Founded DATE
);
```

Script Output | Task completed in 0.031 seconds

```
Table EMPLOYEES created.


Table DEPARTMENTS created.
```

Q5

```
--22p-9326
ALTER TABLE Departments ADD CONSTRAINT unique_department_name UNIQUE (Department_name);
ALTER TABLE Departments ADD CONSTRAINT check_date_founded CHECK (Date_Founded > TO_DATE('01-JAN-2000', 'DD-MON-YYYY'));
```

Script Output | Task completed in 0.094 seconds

```
Table EMPLOYEES created.

Table DEPARTMENTS created.

Table DEPARTMENTS altered.

Table DEPARTMENTS altered.
```

Q6

```
--22p-9326
ALTER TABLE Employees ADD CONSTRAINT pk_employee_id PRIMARY KEY (Employee_id);
ALTER TABLE Departments ADD CONSTRAINT pk_department_id PRIMARY KEY (Department_id);


--22p-9326
```

Script Output | Task completed in 0.079 seconds

```
Table EMPLOYEES altered.

Table DEPARTMENTS altered.
```

Q7

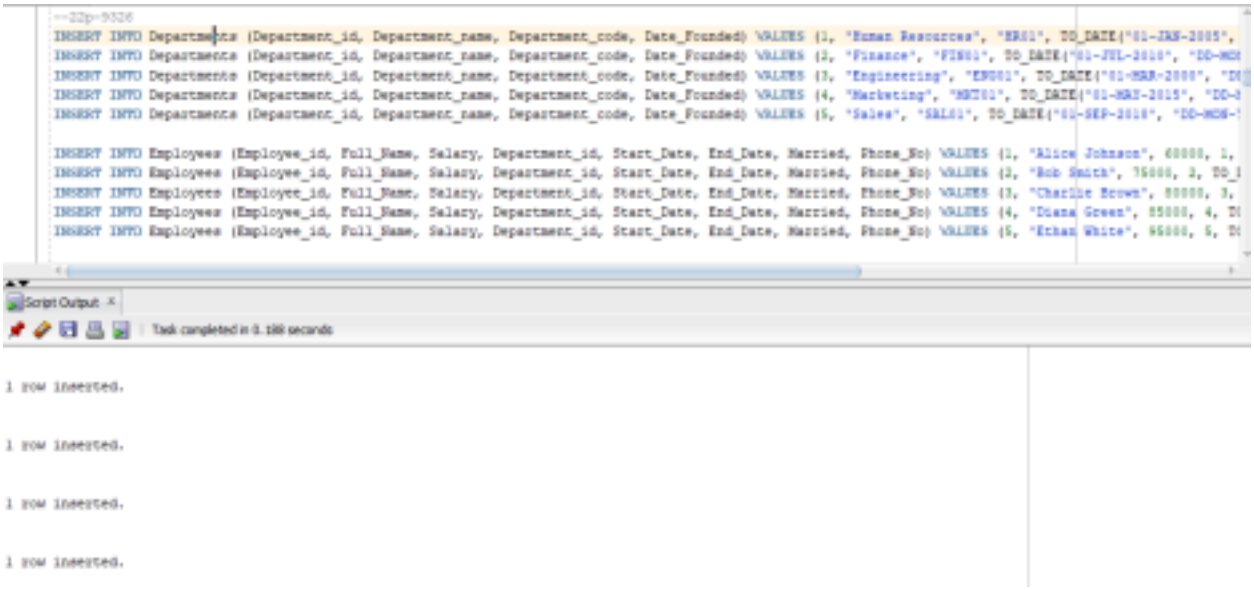
```
--22p-9326

ALTER TABLE Employees ADD CONSTRAINT fk_department_id FOREIGN KEY (Department_id) REFERENCES Departments (Department_id);
```

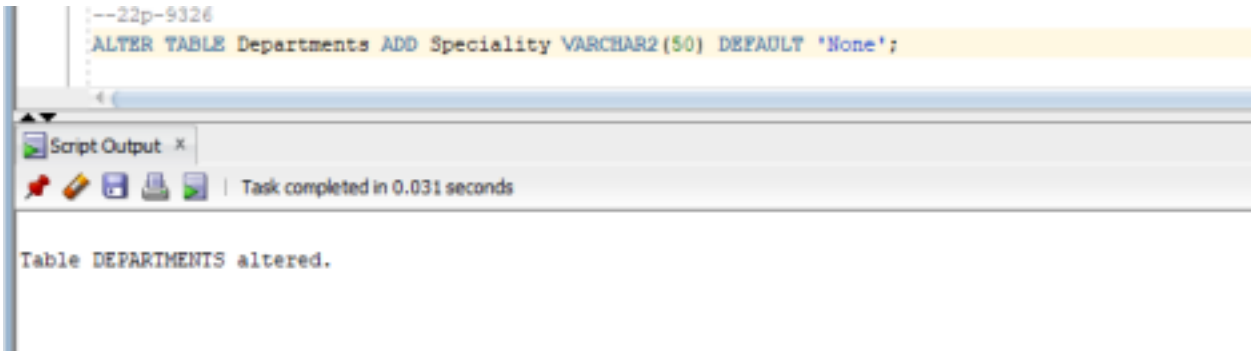
Script Output | Task completed in 0.047 seconds

```
Table EMPLOYEES altered.
```

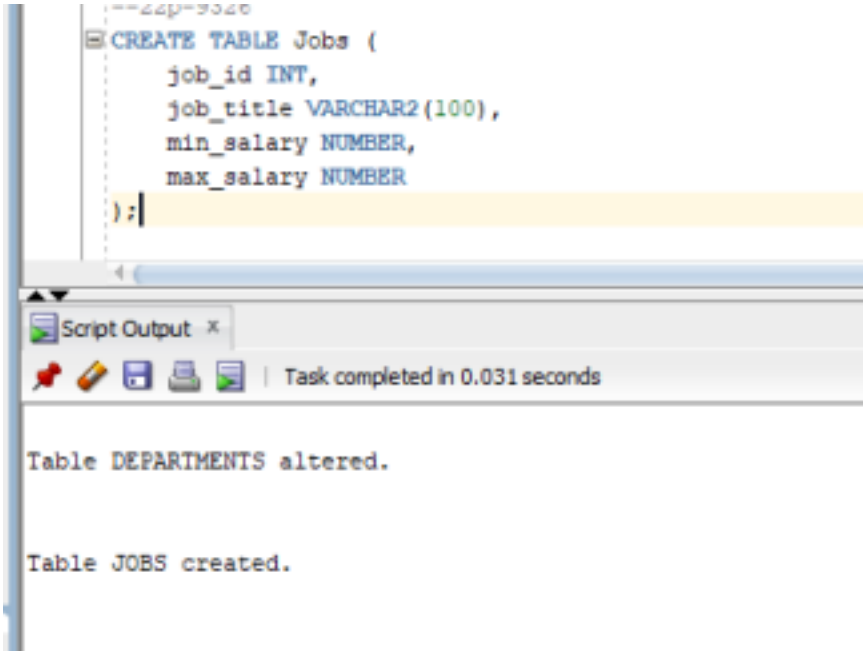
Q8


--22p-9326
INSERT INTO Departments (Department_id, Department_name, Department_code, Date_Founded) VALUES (1, 'Human Resources', 'HR01', TO_DATE('01-JAN-2005',
INSERT INTO Departments (Department_id, Department_name, Department_code, Date_Founded) VALUES (2, 'Finance', 'FIN01', TO_DATE('01-JUL-2010', 'DD-MON
INSERT INTO Departments (Department_id, Department_name, Department_code, Date_Founded) VALUES (3, 'Engineering', 'ENG01', TO_DATE('01-MAR-2000', 'D
INSERT INTO Departments (Department_id, Department_name, Department_code, Date_Founded) VALUES (4, 'Marketing', 'MKT01', TO_DATE('01-MAY-2015', 'DD-
INSERT INTO Departments (Department_id, Department_name, Department_code, Date_Founded) VALUES (5, 'Sales', 'SAL01', TO_DATE('01-SEP-2018', 'DD-MON-
INSERT INTO Employees (Employee_id, Full_Name, Salary, Department_id, Start_Date, End_Date, Married, Phone_No) VALUES (1, 'Alice Johnson', 60000, 1,
INSERT INTO Employees (Employee_id, Full_Name, Salary, Department_id, Start_Date, End_Date, Married, Phone_No) VALUES (2, 'Bob Smith', 75000, 2, TO_
INSERT INTO Employees (Employee_id, Full_Name, Salary, Department_id, Start_Date, End_Date, Married, Phone_No) VALUES (3, 'Charlie Brown', 80000, 3,
INSERT INTO Employees (Employee_id, Full_Name, Salary, Department_id, Start_Date, End_Date, Married, Phone_No) VALUES (4, 'Diana Green', 85000, 4, TO
INSERT INTO Employees (Employee_id, Full_Name, Salary, Department_id, Start_Date, End_Date, Married, Phone_No) VALUES (5, 'Ethan White', 65000, 5, TO
Script Output
Task completed in 0.188 seconds
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.

Q9


--22p-9326
ALTER TABLE Departments ADD Speciality VARCHAR2(50) DEFAULT 'None';
Script Output
Task completed in 0.031 seconds
Table DEPARTMENTS altered.

Q10


--22p-9326
CREATE TABLE Jobs (
job_id INT,
job_title VARCHAR2(100),
min_salary NUMBER,
max_salary NUMBER
);
Script Output
Task completed in 0.031 seconds
Table DEPARTMENTS altered.
Table JOBS created.

Q11

```
--22p-9326
ALTER TABLE Jobs ADD CONSTRAINT pk_job_id PRIMARY KEY (job_id);
```

Script Output

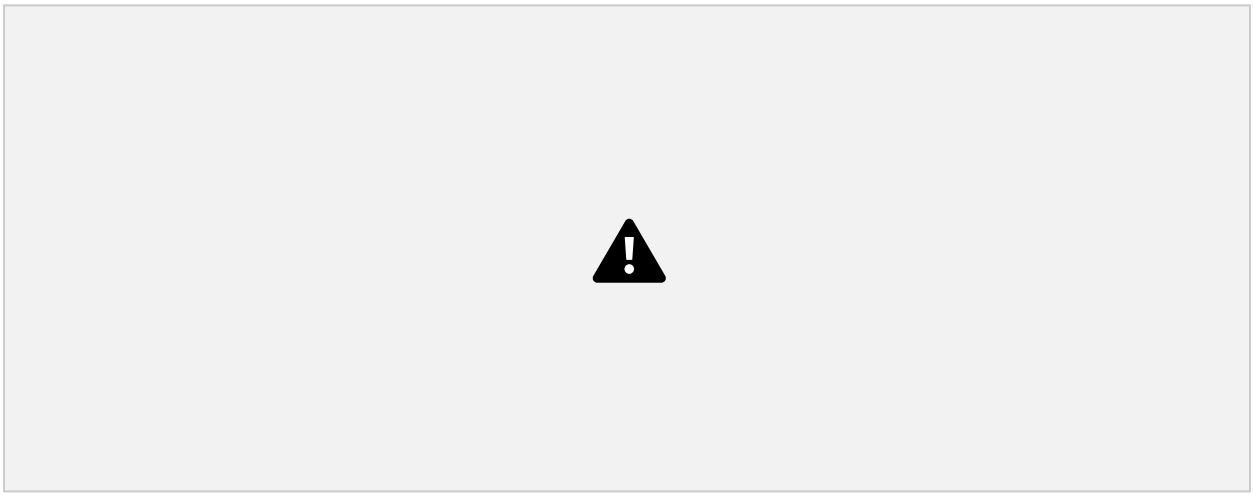
Task completed in 0.032 seconds

```
Table DEPARTMENTS altered.

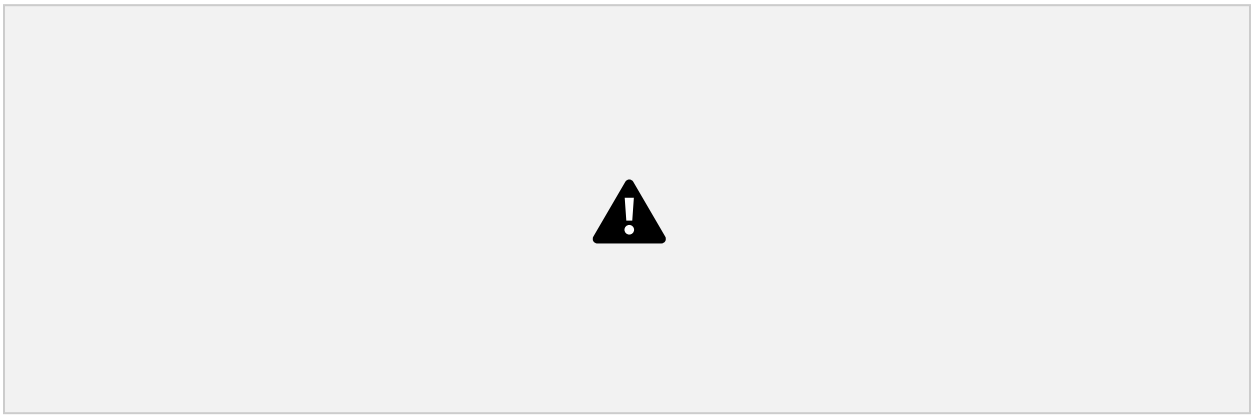

Table JOBS created.


Table JOBS altered.
```

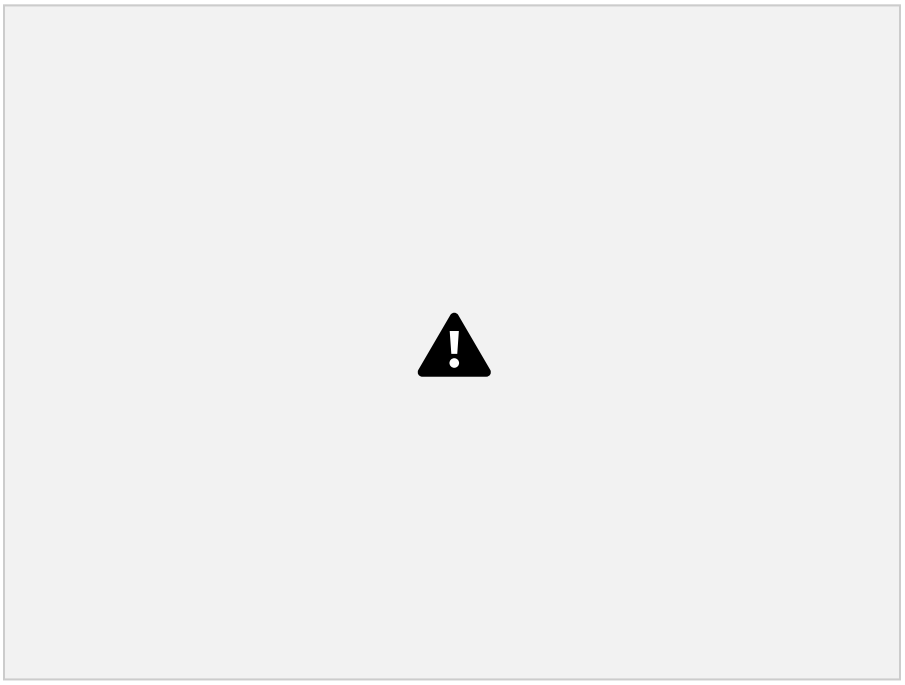
Q12



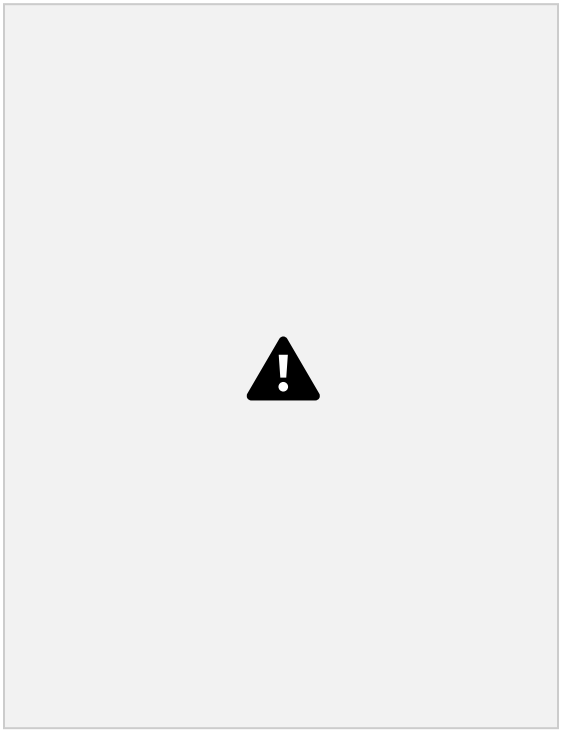
Q13



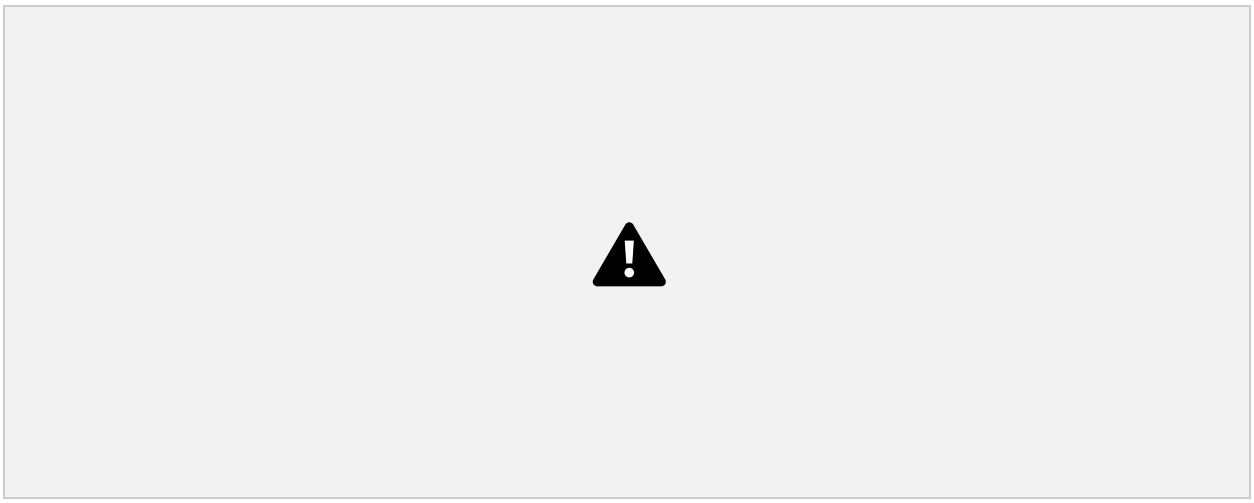
Q14



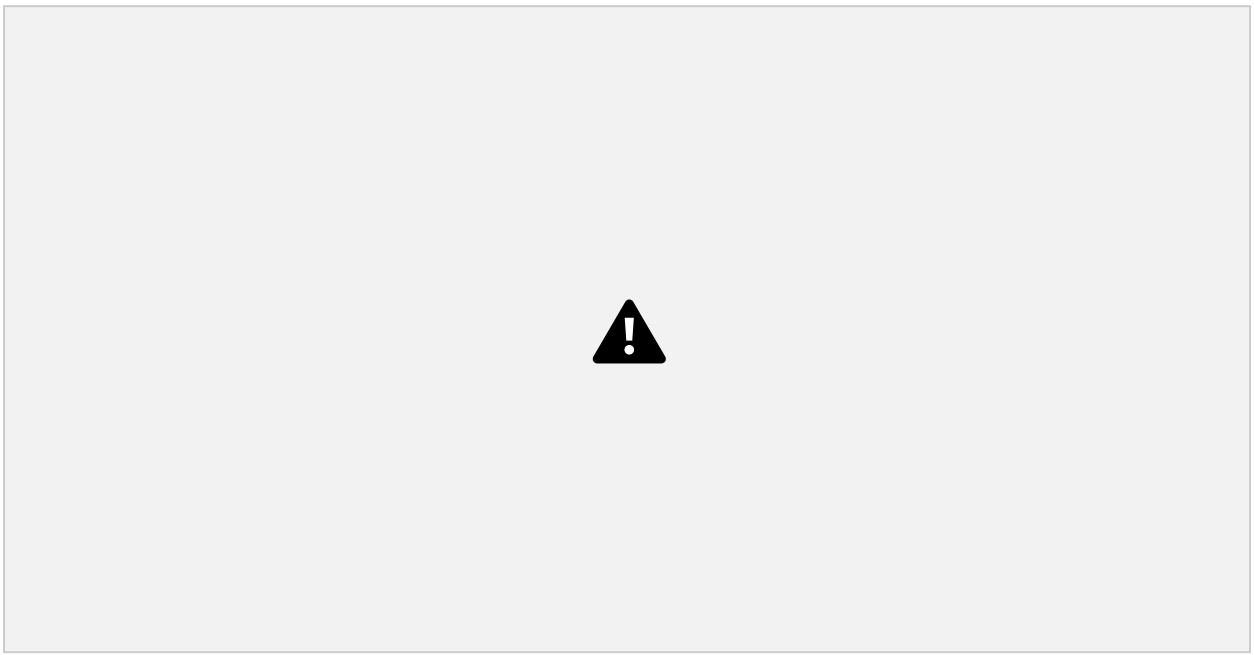
Q15



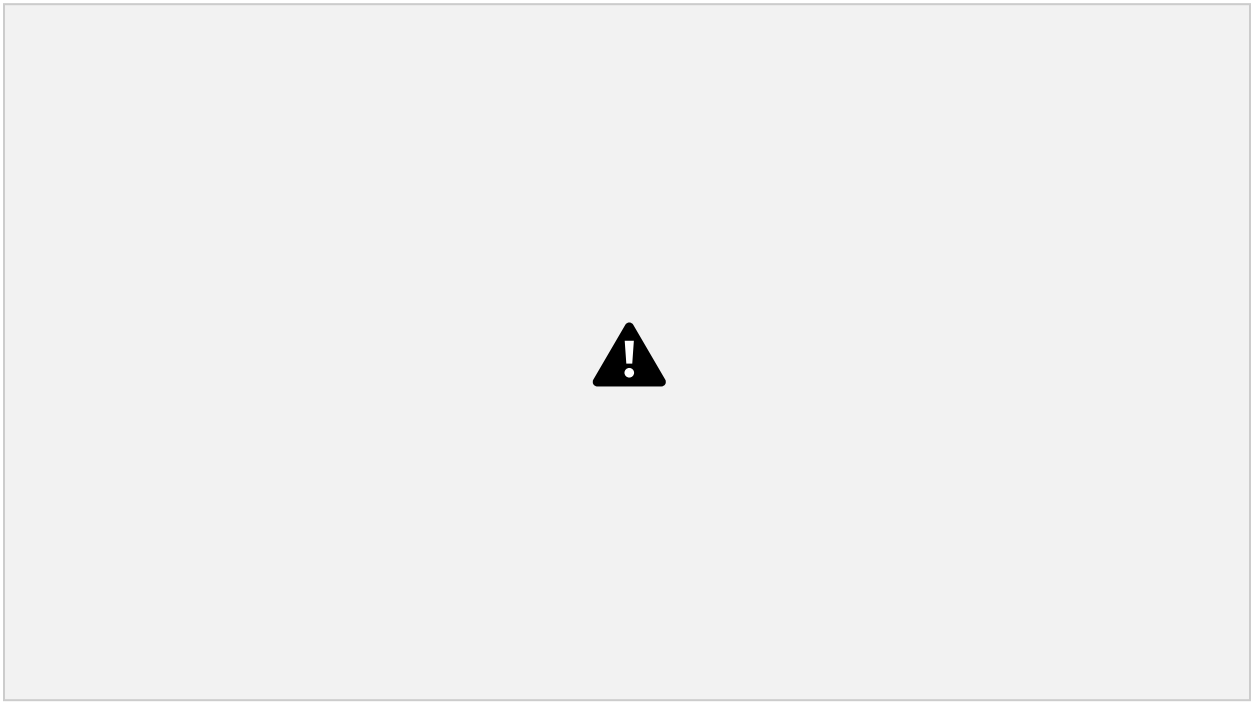
Q16



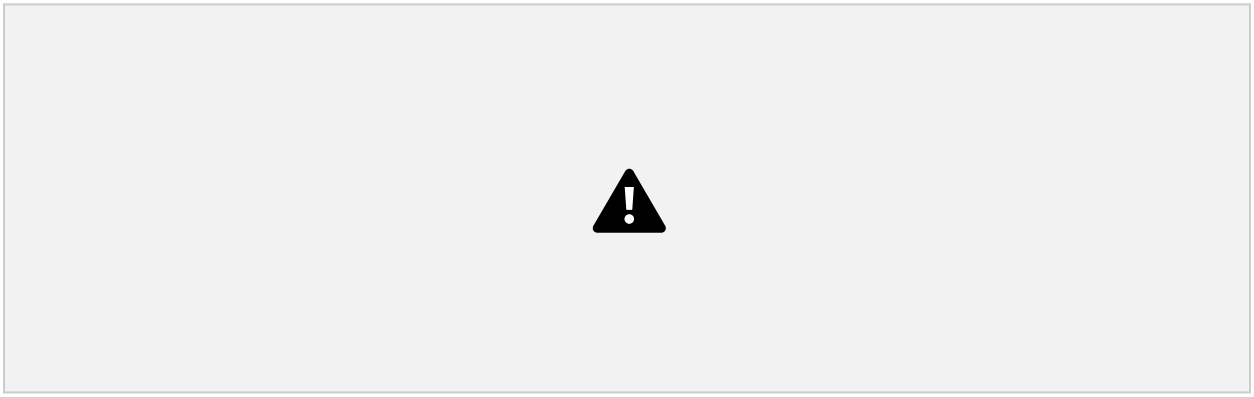
Q17



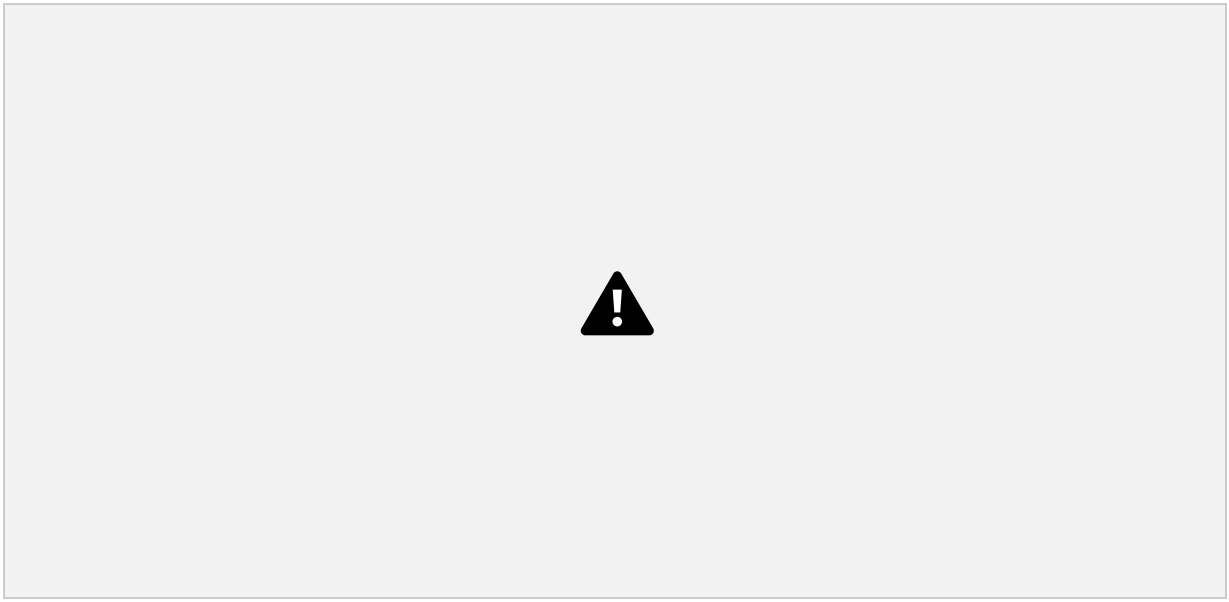
Q18



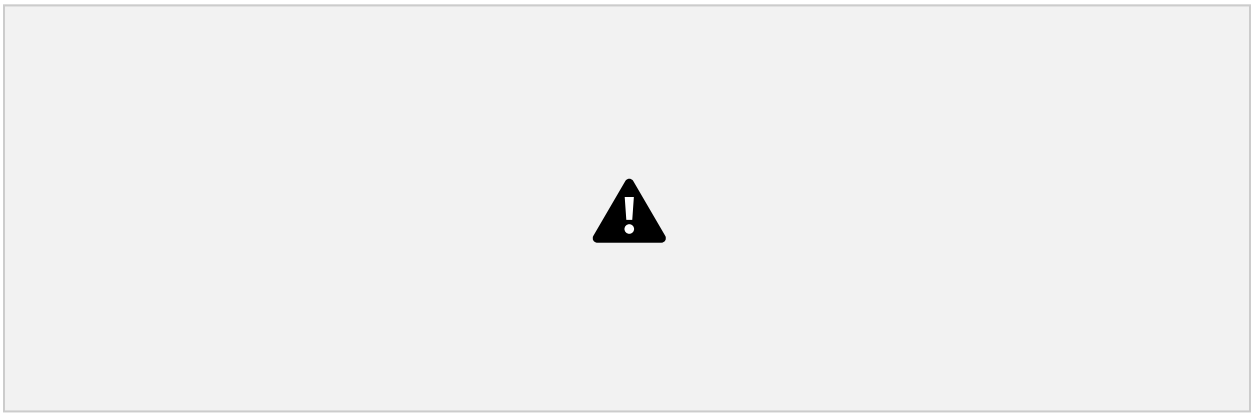
Q19



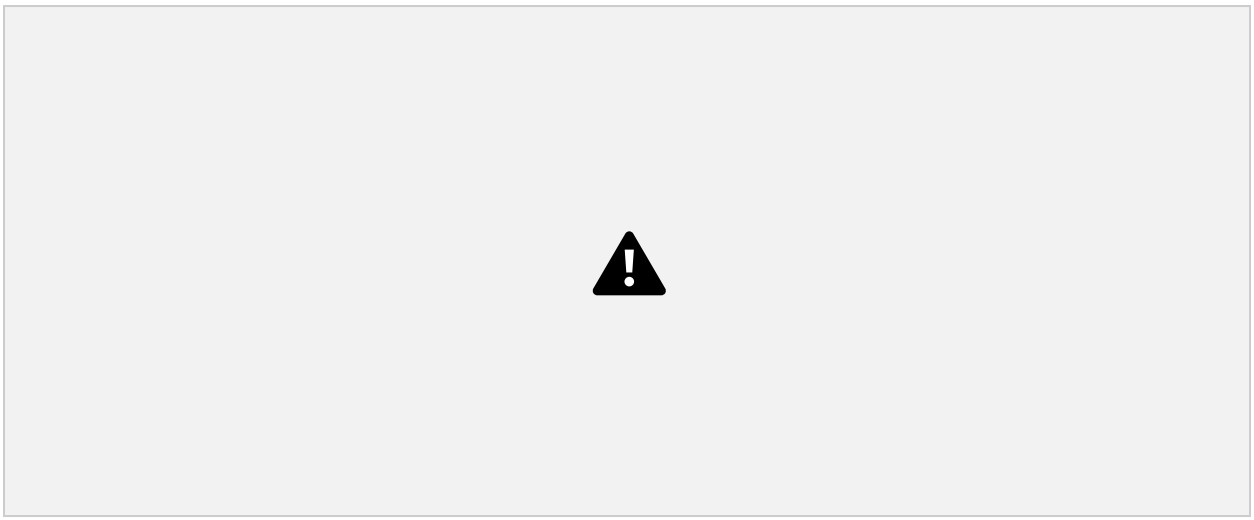
Q20



Q21

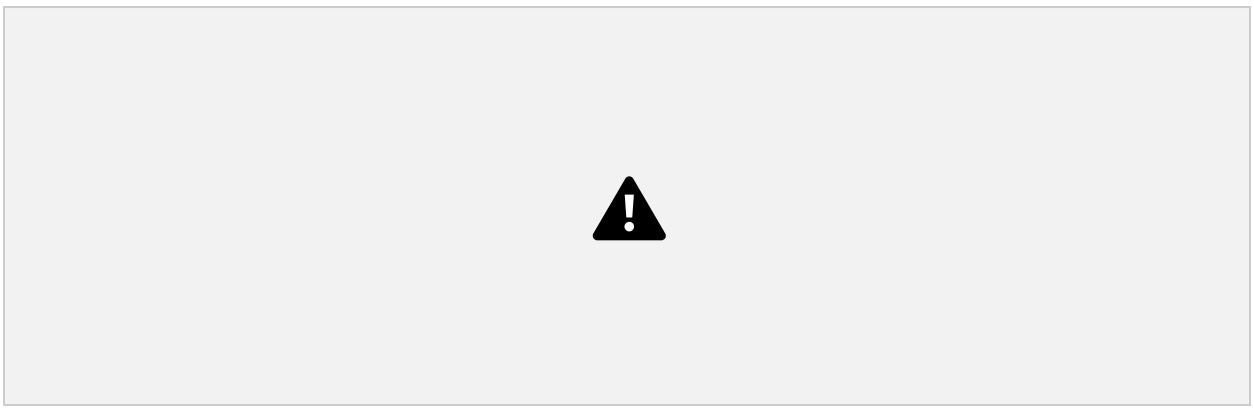


Q22

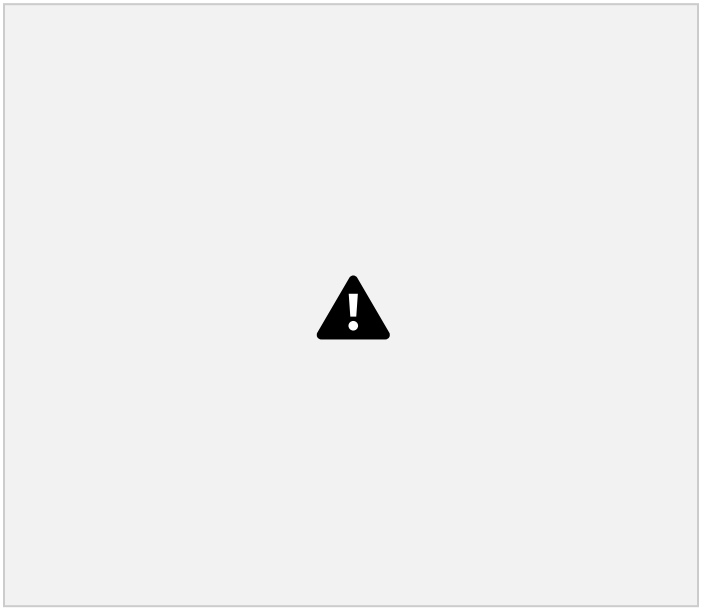


This command will give error because both ids are in different tables

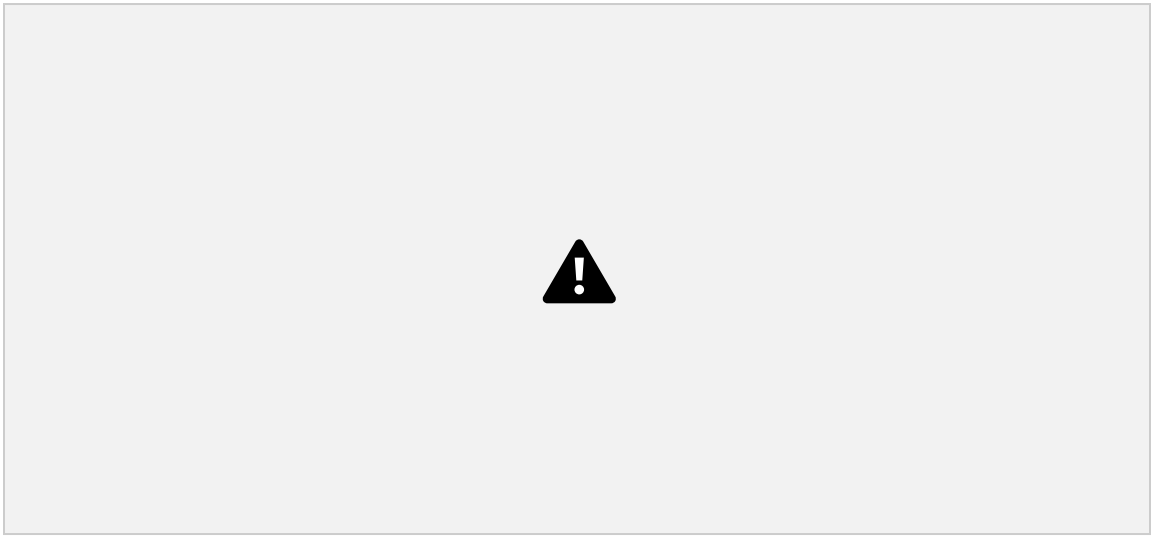
Q23



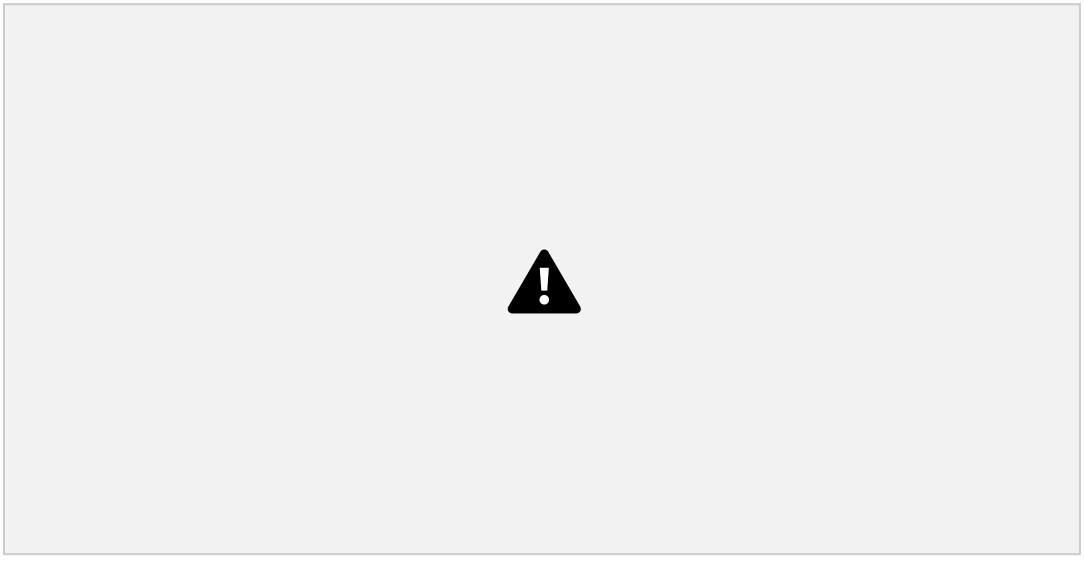
Q24



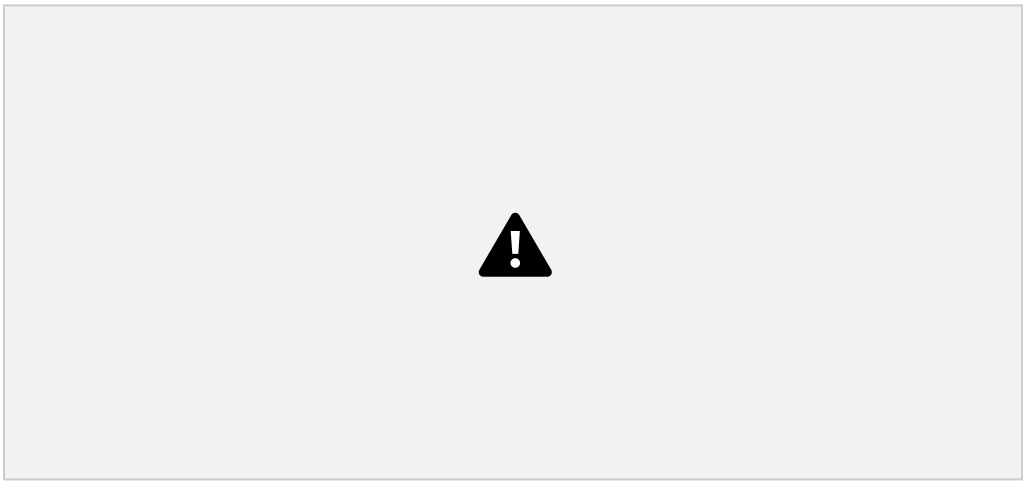
Q25



Q26



Q27



Q28

